Framing a Machine Learning Problem

Module 14 Study Notes • Project Foundation

1. Business-to-ML Translation

The most common mistake in machine learning is failing to translate an abstract business goal into a concrete, measurable technical task.

- **Defining Objectives:** Convert broad goals (e.g., "increase revenue" or "improve user experience") into specific, quantifiable metrics (e.g., "reduce customer churn by 5%" or "increase recommendation click-through-rate by 10%").
- **Problem Type Identification:** Determine the nature of the task. You must decide if the problem is **Supervised** (e.g., Classification or Regression) or **Unsupervised** (e.g., Clustering or Dimensionality Reduction).

2. Scoping and Constraints

Defining the boundaries of the project early helps manage stakeholder expectations and resource allocation.

- **Feasibility Assessment:** Determine if the problem is actually solvable using machine learning or if it could be addressed with simpler, rule-based heuristics.
- **Constraint Estimation:**
  - **Budget:** Calculate the financial resources allocated for data acquisition, infrastructure, and engineering hours.
  - **Timeline:** Set clear deadlines for data collection, model development, testing, and deployment.
- **Production Context:** Decide how the model will function in the real world. Will it operate as a **Batch** process (updated periodically) or an **Online** learning system (adapting in real-time)?

3. Data Strategy

A successful project requires a solid plan for the data that will fuel the model.

The Data Checklist:

- **Availability:** Identify the source of the data (e.g., internal data warehouses, third-party APIs, or web scraping).
- **Sufficiency:** Assess if the existing data is sufficient in both quantity and quality to train a reliable model.
- **Feature Accessibility:** Validate that the necessary input features required to make a prediction will be available at the time of inference.